

Section 1.4 Exercise Answers

1. a. $2x + 6$; b. $2x^2 + 2x - 6$; c. $-x^4 - 2x^3 + 6x^2 + 12x$; d. $\frac{x^2+2x}{6-x^2}$
2. a. $-3x^2 + x + 5$; b. $-3x^2 + x - 5$; c. $-15x^2 + 5x$; d. $\frac{-3x^2+x}{5}$
3. a. $\frac{4x^3+8x^2+1}{2x}$; b. $\frac{4x^3+8x^2-1}{2x}$; c. $x + 2$; d. $4x^3 + 8x^2$
4. a. $-\frac{2}{(x-4)(6-x)}$; b. $\frac{-2x+10}{(x-4)(6-x)}$; c. $-\frac{1}{(x-4)(6-x)}$; d. $\frac{6-x}{x-4}$
5. a. $3x^2 + \sqrt{x-5}$; b. $3x^2 - \sqrt{x-5}$; c. $3x^2\sqrt{x-5}$; d. $\frac{3x^2}{\sqrt{x-5}}$
6. a. 3 b. $18x^2 - 60x + 51$ c. $6x^2 - 2$ d. $9x - 20$ e. 163
7. $f(g(x)) = x + 3$ and $g(f(x)) = \sqrt{x^2 + 3}$
8. $f(g(x)) = \sqrt{x^2 + 3} + 2$ and $g(f(x)) = x + 4\sqrt{x} + 7$
9. $f(g(x)) = |5x + 1|$ and $g(f(x)) = 5|x| + 1$
10. $f(g(x)) = \frac{\sqrt[3]{x+1}}{x}$ and $g(f(x)) = \frac{\sqrt[3]{x+1}}{x}$
11. $f(g(x)) = \frac{x}{7}$ and $g(f(x)) = 7x - 36$
12. $f(g(x)) = \frac{x}{2}$ and $g(f(x)) = 2x - 4$
13. $(\sqrt{x} - 6)^4 + 6$
14. $\frac{1}{(x+3)^2} + 1$
15. a. $\frac{1}{x-3}$ b. $\frac{1-3x}{x}$ c. $\frac{1}{x(x-3)}$
16. $-\frac{3}{\sqrt{2-4x}}$
17. a. $\frac{x^2}{2x^2-2x+1}$ b. $\frac{4}{5}$
18. $f(x) = x^2$; $g(x) = x + 2$
19. $f(x) = x^3$; $g(x) = x - 5$
20. $f(x) = 3x$ $g(x) = \frac{1}{x-5}$
21. $f(x) = \frac{4}{x}$ $g(x) = (x + 2)^2$
22. $f(x) = 4 + x$; $g(x) = \sqrt[3]{x}$
23. $f(x) = \sqrt[3]{x}$; $g(x) = \frac{1}{2x-3}$
24. $f(x) = x^{-3}$; $g(x) = \frac{1}{3x^2-4}$
25. $f(x) = \sqrt[4]{x}$; $g(x) = \frac{3x-2}{x+5}$
26. $f(x) = x^4$; $g(x) = \frac{8+x^3}{8-x^3}$
27. $f(x) = \sqrt{x}$; $g(x) = 2x + 6$
28. $f(x) = x^3$; $g(x) = 5x - 1$
29. $f(x) = \sqrt[3]{x}$; $g(x) = x - 1$
30. $f(x) = |x|$; $g(x) = x^2 + 7$
31. $f(x) = x^3$; $g(x) = \frac{1}{x-2}$
32. $f(x) = \frac{1}{x^2}$; $g(x) = 2x - 3$
33. $f(x) = \sqrt{x}$; $g(x) = \frac{2x-1}{3x+4}$
34. 0
35. 2
36. 4
37. 5

- 38. 3
- 39. 4
- 40. 2
- 41. 0
- 42. 1
- 43. 2
- 44. 4
- 45. 1
- 46. 4
- 47. 4
- 48. 4
- 49. 4
- 50. 4
- 51. 9
- 52. 9
- 53. 4
- 54. 4
- 55. 2
- 56. 7
- 57. 3
- 58. 5
- 59. 11
- 60. 0
- 61. 0
- 62. 1
- 63. 7
- 64. -57
- 65. -94
- 66. 4
- 67. 5
- 68. 243
- 69. $18x^2 + 60x + 51$
- 70. 62
- 71. $9x + 20$